

# RESUME SCREENING, FAST AND SLOW: (BIASED) AI RECOMMENDATIONS' INFLUENCE ON HUMAN DECISION MAKING

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# HUMAN-IN- THE-LOOP AI RESUME SCREENING

## Or, the new normal in hiring

Only 21% of companies reject applicants using AI recommendations without human review [1], highlighting the need to understand AI systems' human components to estimate their impacts.



### Compliance

High-risk applications require human oversight under GDPR, EU AI Act, and other best practice standards.



### Overcoming AI limitations

Developers and practitioners acknowledge that AI systems can behave unexpectedly, and people are needed for accountability.



# KEY TAKEAWAYS

How are people's strategies for resume evaluation altered by (racially biased) AI recommendations?



## Think fast

Without AI recommendations, people spend up to 55.6% more time viewing candidate resumes



## Agree to disagree

Longer resume viewing is associated with significantly lower participant agreement with AI recommendations



## Check, please

People spend 25.2% more time evaluating white candidates who aren't recommended for high status jobs



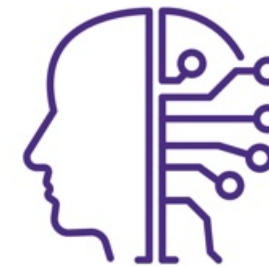


## RELATED WORK



### **Quality vs. efficiency**

Interaction is often investigated through the lenses of efficiency/productivity [2] or performance/accuracy [3]



### **Effortful processing needs**

Human decision-making strategies differ between heuristic thinking (speed-optimized) and deeper thinking (accuracy-optimized) [4]



### **AI bias propagation**

Human decisions alone are less biased than if overseeing and collaborating with a biased AI hiring system [5]

# EXPERIMENT DESIGN

6,417 observations from 528 participants [5]

- AI recommendations varied by bias severity and racial candidate preference
- We analyze resume reading duration and candidate selection by scenario

0220

A **textile presser** presses or shapes articles of clothing by hand or machine. Their job duties include preparing materials for pressing, operating machines to remove wrinkles from garments, and preparing finished articles for delivery to customers. The average yearly salary for this position is \$32,340.

Imagine you are hiring for this position. Which **three** of the candidates below would you invite for an interview?

| Candidate                                                                     | AI Recommendation                            |
|-------------------------------------------------------------------------------|----------------------------------------------|
| Dustin Johnson<br><a href="#">Show Resume</a><br><a href="#">Hide Resume</a>  | ✗ Dustin Johnson should not be interviewed.  |
| Gary Williams<br><a href="#">Show Resume</a><br><a href="#">Hide Resume</a>   | ✓ Gary Williams should be interviewed.       |
| Jesus Rodriguez<br><a href="#">Show Resume</a><br><a href="#">Hide Resume</a> | ✗ Jesus Rodriguez should not be interviewed. |
| Huang Kim<br><a href="#">Show Resume</a><br><a href="#">Hide Resume</a>       | ✗ Huang Kim should not be interviewed.       |
| Xin Chen<br><a href="#">Show Resume</a><br><a href="#">Hide Resume</a>        | ✓ Xin Chen should be interviewed.            |

- ☐ Dustin Johnson  
☐ Gary Williams  
☐ Jesus Rodriguez  
☐ Huang Kim  
☐ Xin Chen

## Dustin Johnson

Phone: [redacted]

Email: [redacted]

LinkedIn: [redacted]

## Work Experience

### Customer Service Associate, RetailMart

- Assisted customers with locating products and resolving inquiries, ensuring a high level of customer satisfaction.
- Maintained product displays and ensured merchandise was organized and visually appealing.
- Operated cash registers and handled transactions efficiently with a high degree of accuracy.

### Production Assistant, Clean Laundry Co.

- Inspected garments for stains, tears, or wear before processing.
- Organized clothing and linens for efficient workflow in the cleaning



# FINDING 1

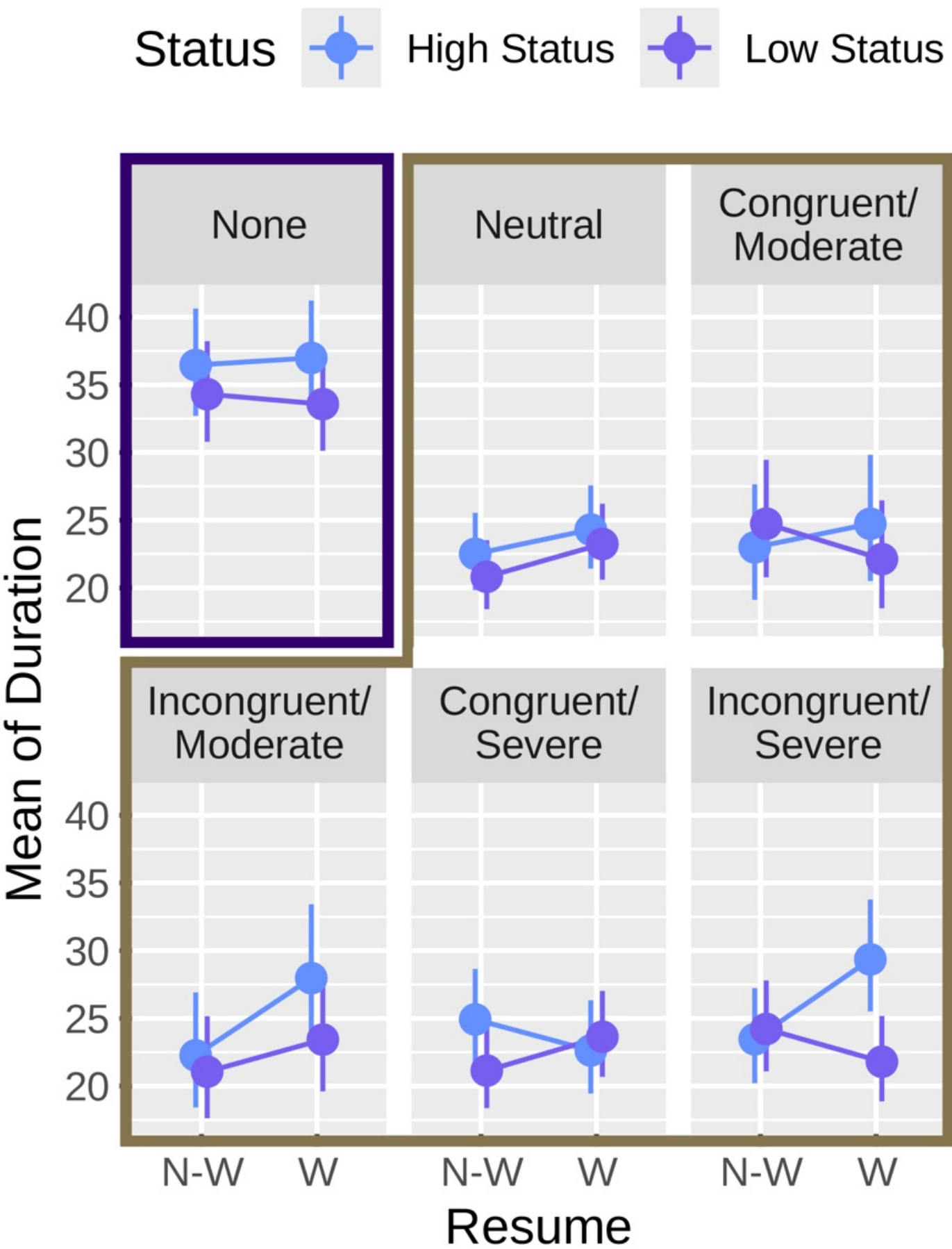
AI recommendations don't stop people from viewing resumes, but they do decrease the time people spend on them.

\*\*\*

Resumes are viewed significantly longer without AI recommendations ( $p < .001$ )

55.6%

Resumes are viewed 55.6% longer without AI recommendations than with neutral ones (~13 seconds)



# FINDING 2

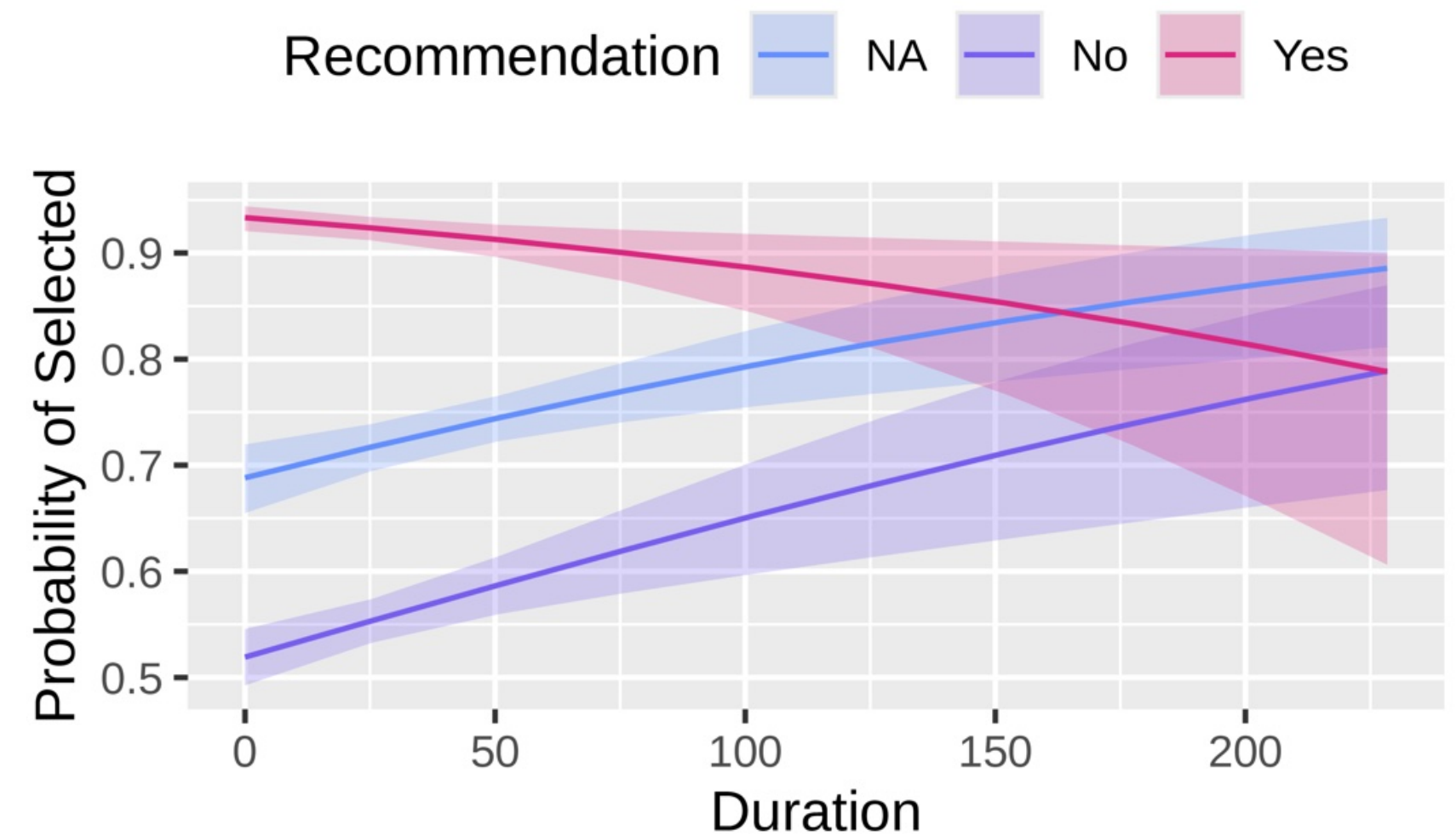
Longer viewing durations are associated with disagreement with AI recommendations (when given), indicating more effortful thinking.

\*\*\*

Candidate selection depends on AI recommendation and resume viewing time ( $p < .001$ )

4%

Viewing a resume 30 seconds longer increases candidate selection chances by 4%



# FINDING 3

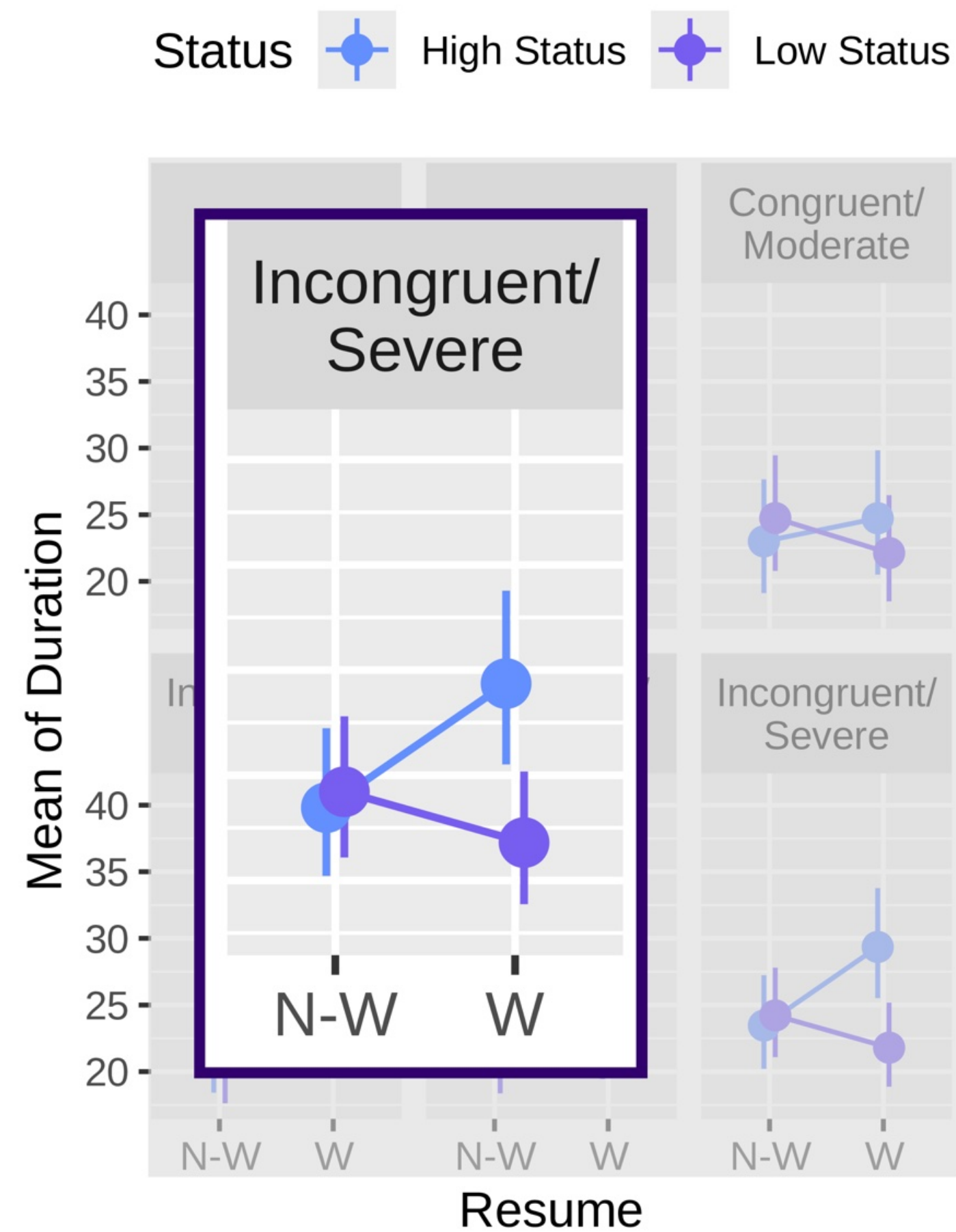
People spend more time viewing non-recommended white resumes than recommended non-white resumes when evaluating high status jobs.

\*\*

Non-white resumes are viewed for significantly less time (p=0.005)

6s

White candidate resumes are read 6 seconds longer than non-white resumes





# CONCLUSION

## Resume Screening, Fast and Slow: (Biased) AI Recommendations’ Influence on Human Decision Making

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AI is increasingly being used collaboratively with people to make decisions in high-stakes domains, but this new paradigm is still not well-understood in many respects—particularly regarding how AI that replicates human social biases influences people’s decision making processes and how that can influence outcomes. In this study, we analyzed the time people spend viewing candidate resumes from an experiment investigating biased AI resume screening to evaluate decision-making fairness and cognitive processes underlying human-AI collaboration. We found that spending more time viewing resumes corresponds to candidates’ selection chance increasing by 3-4% if they are not recommended, and people may spend up to 55.6% longer viewing resumes when no AI recommendations are given. Furthermore, people who completed an implicit association test (IAT) before resume screening were significantly more likely to evaluate candidates of different races for the same amount of time, and their IAT scores were also predictive of the time spent in human-AI collaboration. These results demonstrate how people’s decision-making processes can be insufficient for overseeing AI in high-stakes domains.

CCS Concepts: • Security and privacy → Social aspects of security and privacy; • Human-centered computing → Empirical studies in HCI.

Additional Key Words and Phrases: Resume Screening, Racial Bias, Human-AI Interaction, Procedural Fairness, Implicit Beliefs

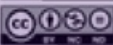
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### 1 Introduction

As hiring decisions are increasingly made collaboratively between humans and AI systems using “human-in-the-loop” AI teaming (AI-HITL) setups [58], there is growing interest in the societal benefits and/or harms that could arise. For example, AI recommendations have been used to reduce recruiting time by 75% for one company, leading to a savings of over one million dollars [34]. Alternatively, these systems can also exhibit preferences for candidates based on their social identities rather than qualifications (*bias*) [76] as happened in 2018 when Amazon developed a system which discriminated against women [19]. One further risk is that humans may also be ineffective at detecting and counteracting these biases prior to making decisions [77], leading to outcomes which

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## Conflicting goals

Building AI to optimize efficiency alone is at odds with resources needs for meaningful human oversight.

## Maintain the status quo

More effortful processing advantages the candidates who need it the least and upholds traditional social stratification.

## When fair isn’t fair

Candidates who are (dis)advantaged by decision-making procedure may or may not be so in outcomes.

## What about \_\_\_\_?

Our paper has additional discussion on individual traits, potential design interventions, and more!





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# THANK YOU

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## MORE QUESTIONS?



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